

positive voltage regulator, capable of supplying in excess of 5 A over a 1.2V-32V output range. It is exceptionally

easy to use and requires only two resistors to set the output voltage.

A unique feature of LM338 is time-dependent current limiting. The current-limiting circuitry allows peak currents of up to 12 A to be drawn from the regulator for short periods of time. This allows LM338 to be used with heavy transient loads and speeds startup under full-load conditions. Under sustained loading conditions, the current limit decreases to a safe

if the adjustment pin is accidentally disconnected.

7805 and 7812 are three-terminal positive voltage regulators, available in TO-220 package with fixed output voltages of 5 V and 12 V, respectively, making them useful in a wide range of applications. Each type employs internal current limiting, thermal shutdown and safe operating area protection. If adequate heat-sinking is provided, these regulators can deliver over 1A output current. A maximum of 35V DC can be supplied to the input of these regulators, provided the device is properly mounted on a heat-sink.

The 20V AC secondary output of the transformer (X1) is directly fed to the bridge rectifiers, which rectify the AC component and convert it into DC voltage.

Since we are using a bridge rectification configuration, the ideal average magnitude of the bridge rectifier output is given as:

$$V_{\text{dc(ideal)}} = V_{\text{avg}} = V_{\text{in}} \times \sqrt{2} = 20 \times \sqrt{2} = 28.2848 \text{ volts}$$

After subtracting the voltage drop across the two diodes that work in series in bridge rectification, we get the actual approx. output voltage as:

$$V_{\text{dc(actual)}} = 28.2848 - (2 \times 0.7) = 26.88 \text{ volts or approx. 27 volts}$$

This voltage output is filtered with the help of capacitor C5, which acts as a low-pass filter. LED1 lights up when the circuit is switched on.

Then this signal is fed to the input of the three regulators (IC2 through IC4).

As per the datasheet of LM338, a drop of 1.5 to 3 volts appears in the output of the regulator when the working temperature increases from -75°C to +150°C. However, at normal working range of 20°C to 40°C, a typical drop of 2.6 V is obtained when the current drawn from the device is maximum 5 amperes. Thus at practical 27V DC input, we obtain a variable voltage range of 1.2 V to 24.4 V, which is sufficient for

Test Points for Power Supply Section

Test point	Details
TP0	0V, GND
TP1	12V
TP2	5V
TP3	1.2V to 25V

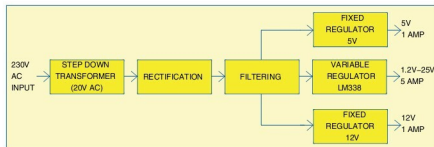


Fig. 2: The main power stage block diagram

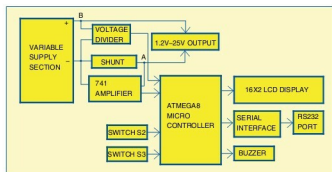


Fig. 3: The interface and display block diagram

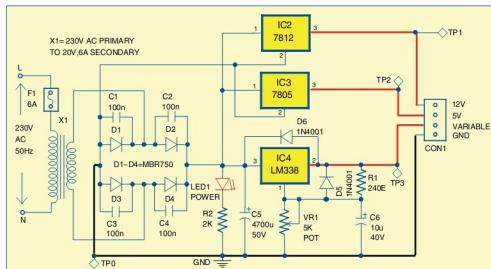


Fig. 4: The main power stage circuit